

Notes: Liberti and Mian (RFS, 2009)
Estimating the Effect of Hierarchies on Information Use

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1 Big picture

- **Question.** Do hierarchies affect how information is used inside firms? More specifically, when the loan officer who collects information is farther from the officer who approves the loan, does the organization rely less on subjective information and more on objective information?
- **Setting.** The paper studies the corporate loan approval process of a large multinational bank in Argentina in 1998. The sample contains 424 corporate loan applicants, and the data come from detailed credit dossiers.
- **Core institutional feature.** A loan officer always collects the information, but final credit approval can occur at one of five hierarchical levels. Some loans are approved by the loan officer or a nearby superior, while others are sent farther up the hierarchy.
- **Key information distinction.** Objective information is based on audited financials, size, leverage, profitability, cash flows, and related verifiable measures. Subjective information is based on the loan officer's qualitative assessment of management quality, competitive position, risk management, industry outlook, and access to capital.
- **Main prediction.** Subjective information is harder to communicate, verify, and use across hierarchical distance. Therefore, when approval occurs farther from the information-collecting loan officer, credit decisions should become less sensitive to subjective information and more sensitive to objective information.
- **Main empirical result.** The data strongly support this prediction. At higher approval levels, the coefficient on subjective information falls sharply, while the coefficient on objective information rises sharply.
- **Mechanism clue.** The sharp change occurs when the approving officer is geographically separated from the loan officer. This suggests that face-to-face interaction, repeated contact, and human touch are important for transmitting soft information.
- **Additional mechanism clue.** The loss of subjective-information use is weaker when the loan officer is more experienced. More experienced loan officers may be more trusted or better understood by senior approvers.
- **Economic takeaway.** The design of hierarchy changes the type of information used in decision making. Layers do not merely allocate authority; they also reshape what the organization can credibly communicate and act on.

2 Incremental contribution of the paper

- **From theory to direct intrafirm evidence.** Theoretical work on organizations predicts that hierarchy makes subjective information harder to use, but direct empirical evidence is rare because internal information flows are usually unobserved. This paper observes information collection, information type, approval authority, and final loan limits within one organization.
- **A direct measure of subjective versus objective information.** Instead of inferring soft information indirectly, the paper uses bank credit files that separately record objective financial information and subjective loan-officer assessments.

- **A clean organizational-design variable.** The paper observes the hierarchical level that approves each loan. This provides a concrete empirical counterpart to “hierarchical distance.”
- **Rule-based quasi-exogenous variation.** The bank’s credit manual assigns applications to approval levels using predetermined observable rules. The paper exploits nonlinearities and discontinuities in these rules to separate the effect of hierarchy from the types of firms sent up the hierarchy.
- **Loan-officer fixed effects.** Because the same loan officer may collect information for loans approved at different levels, the paper can compare loans handled by the same information collector. This helps rule out the idea that high-level and low-level loans simply have different-quality loan officers.
- **Geographical-distance insight.** The paper shows that hierarchical distance matters most when it also creates physical separation. This connects organizational hierarchy to communication frictions.
- **Human-touch contribution.** The results show that experience and proximity mitigate the loss of subjective information, suggesting that trust, repeated interactions, and shared context help transmit soft information.
- **Broader contribution.** The paper gives empirical content to theories of authority, communication costs, incentive effects, and the limits of transmitting nonverifiable information within firms.

3 Data and measurement

3.1 Bank, borrowers, and loan applications

- The data cover the credit folders of all 424 corporate clients of a large multinational bank in Argentina in 1998.
- A firm is classified as corporate by the bank if annual net sales exceed 50 million pesos.
- The credit folder records the full lifecycle of a loan approval: initial application, information collected by the loan officer, approval level, location of the approving officer, requested amount, and approved amount.
- The main outcome is the logarithm of the approved credit line.
- Average approved credit is large, and the sample contains meaningful variation across firms in requested credit, size, profitability, assets, and approval level.

Interpretation: The data are unusually rich because they observe the internal information that flows through the bank’s hierarchy, not just final lending outcomes.

3.2 Approval levels and rule variables

- The bank has five approval levels. Level 1 is the loan officer; higher levels are progressively more senior officers.

- The final approval level is determined before the loan officer completes the detailed information review.
- The assignment is based on a credit manual that uses objective, verifiable rule variables.
- The eight primary rule variables are maturity/collateral score, applicant loan size, central bank credit score, foreign bank branch guarantee, family-company indicator, product-market industry benchmark, financial-risk industry benchmark, and declining-industry indicator.
- There are also 11 rare exception variables, such as covenant violations, adverse changes in risk profile, and other risk events.
- Larger, riskier, more complex, more volatile, and less secured applicants are more likely to be sent to higher levels.

Interpretation: The approval assignment process is not random, but it is rule based and observable. This is the foundation for the paper’s identification strategy.

3.3 Objective information

- Objective information is built from audited financial statements and firm-size measures.
- The performance index includes ratios such as interest coverage, funds-flow interest coverage, funds from operations over total debt, free operating cash flow over total debt, pretax return on average capital, total debt over capitalization, and current ratio.
- The size index is based on firm size.
- The bank maps raw ratios into prespecified ratings and constructs an objective index. The authors standardize the index.

Interpretation: Objective information is verifiable and easy to transmit. In theory, it should survive movement up the hierarchy better than subjective information.

3.4 Subjective information

- Subjective information comes from the loan officer’s assessments after interviews, visits, and relationship work.
- It includes industry risk assessment, competitive position, management quality, risk management policies, and access to capital.
- The loan officer assigns ordinal scores, generally from 1 to 7, with higher values indicating better firm quality.
- The paper uses the bank’s aggregate subjective index and also separates it into a strong subjective index and weak subjective index.
- Strong subjective components include management quality and competitive position, where judgment and interpretation are especially important.
- Weak subjective components include industry risk, risk management policy, and access to capital, which are partly informed by more verifiable evidence.

Interpretation: The distinction between strong and weak subjective information allows the authors to test whether the hierarchy effect is stronger precisely when information is harder to verify and transmit.

3.5 Outcome and control variables

- The main outcome is $\log(\text{approved credit})$.
- The paper also observes total facilities, outstanding credit, sales, income, assets, loan-officer identity, approval level, geographical location, and loan-officer tenure.
- The main controls include rule variables, interactions of rule variables with information measures, loan-officer fixed effects, and interactions of loan-officer fixed effects with information measures.

4 Models and empirical specifications

4.1 Information-use prediction

The baseline idea is that a loan amount should respond to both objective and subjective information. If the loan officer approves the loan, both types of information are used close to the source. If a manager approves the loan, subjective information may lose precision, credibility, or salience.

A simplified version of the empirical model is

$$L_{ij} = \alpha + \beta_1 MGR_{ij} + \beta_H H_{ij} + \beta_H^M (H_{ij} \times MGR_{ij}) + \beta_S S_{ij} + \beta_S^M (S_{ij} \times MGR_{ij}) + \varepsilon_{ij},$$

where:

- L_{ij} is log approved credit for firm i , whose information is collected by loan officer j .
- H_{ij} is objective information.
- S_{ij} is subjective information.
- MGR_{ij} indicates approval by a higher-level manager rather than the loan officer.

Prediction:

$$\beta_H^M > 0 \quad \text{and} \quad \beta_S^M < 0.$$

That is, higher-level approval should make credit more sensitive to objective information and less sensitive to subjective information.

4.2 Identification with bank assignment rules

A central concern is that firms sent higher in the hierarchy are different. For example, larger or more complex firms may naturally have more informative audited financials and less informative subjective assessments. The paper addresses this by controlling for the variables used by the bank's credit manual to assign approval levels.

Let Z_i denote the rule variables. The paper includes these variables and their interactions with both information measures. The identification then comes from nonlinearities, thresholds, and discontinuities in the mapping from Z_i to approval level, rather than from smooth differences in firm type.

4.3 Loan-officer fixed effects

A second concern is that some loan officers may be better at collecting subjective information. If better loan officers handle low-level approvals, then the low-level sensitivity to subjective information could reflect loan-officer quality rather than hierarchy.

The richer specification includes loan-officer fixed effects and loan-officer-specific slopes on objective and subjective information:

$$\begin{aligned} L_{ij} = & \beta_H^M(H_{ij} \times MGR_{ij}) + \beta_S^M(S_{ij} \times MGR_{ij}) \\ & + \alpha_j + \alpha_j H_{ij} + \alpha_j S_{ij} + \beta_1 MGR_{ij} \\ & + \beta_2 Z_i + \beta_3(H_{ij} Z_i) + \beta_4(S_{ij} Z_i) + \varepsilon_{ij}. \end{aligned}$$

Interpretation: This compares loans whose information was collected by the same loan officer but approved at different levels.

4.4 Geographical distance, experience, and subjectivity tests

The paper then asks what explains the hierarchy effect.

- **Geographical distance.** The main break occurs when approving officers are not in the same branch as the loan officer. This suggests a communication mechanism.
- **Loan-officer experience.** The decline in subjective-information use is smaller when the information comes from an experienced loan officer. Experience may generate trust, shared language, and repeated interaction.
- **Strong versus weak subjective information.** The hierarchy effect is larger for more subjective subcomponents. This supports the interpretation that the key friction is the difficulty of transmitting nonverifiable information.

5 Identification and empirical design

5.1 What the design can identify

- The paper identifies how the sensitivity of credit approval to different types of information changes with hierarchical and geographical distance.
- The design is not a randomized experiment, but the rule-based assignment process provides plausibly exogenous residual variation after controlling for rule variables and their interactions.
- The strongest claim is that hierarchy changes information use, not just that higher-level loans are different.

5.2 Endogenous bank assignment

- Concern: the bank may deliberately send firms with less useful subjective information or more useful objective information to higher approval levels.
- Response: the bank’s assignment rules are observable and predetermined in credit manuals.
- The paper controls flexibly for rule variables and interactions with objective and subjective information.
- Residual identification comes from nonlinear rule cutoffs rather than from a simple correlation between firm type and approval level.

5.3 Endogenous loan-officer assignment

- Concern: better loan officers may collect information for loans that remain at lower levels.
- Response: the authors use loan-officer fixed effects and interactions between loan-officer fixed effects and information variables.
- This forces comparisons within the same information-collecting officer.

5.4 Interpretation and limitations

- The results are consistent with communication costs, reduced incentives to collect soft information when authority is remote, and loss of credibility of subjective assessments across layers.
- The results are less consistent with pure taste differences across managers, because the effects line up sharply with geography and information subjectivity.
- The paper does not fully distinguish among all theoretical mechanisms: communication costs and incentive effects can both generate reduced use of subjective information.
- External validity is limited because the sample comes from one bank, one country, one year, and approved credit applications.

6 Tables and figures

6.1 Figure 1 and Figure 2: hierarchy and approval process

Read:

- Figure 1 presents the conceptual hierarchy: the loan officer collects subjective and objective information, but the loan decision may be made either by the loan officer or by the manager above him.
- Subjective information is represented as losing informativeness when it travels upward; objective information is represented as preserving informativeness.
- Figure 2 maps the bank’s approval process into five levels. The loan officer collects all information, while credit approval is assigned to one of the five levels by the bank’s credit manual.

Economic message: The figures summarize the paper’s core mechanism. The same information may be collected at the bottom of the organization, but its use depends on how far the decision authority sits from that source.

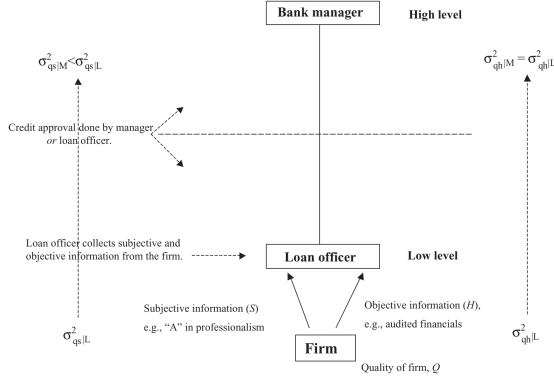


Figure 1
An example of bank hierarchical structure

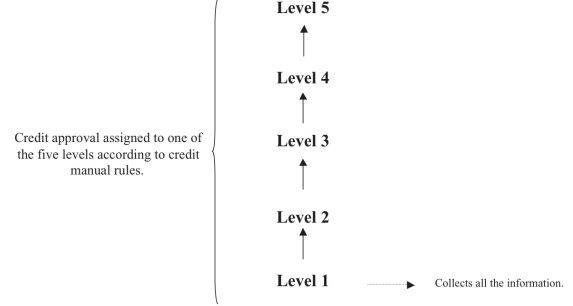


Figure 2
Hierarchical decision-making process

(a) Figure 1: bank hierarchy and information flow

(b) Figure 2: five-level approval process

6.2 Table 1 and Table 2: sample structure and assignment rules

Read:

- Table 1 shows that loans are spread across five approval levels: 26.7% at Level 1, 37.3% at Level 2, and the rest across Levels 3–5.
- Higher approval levels are associated with larger applicant size, worse central-bank credit score, weaker industry benchmarks, and more complex credit requests.
- The objective and subjective indices both vary across approval levels, but there remains substantial within-level variation.
- Table 2 predicts approval level using rule variables. The key rule variables explain a meaningful share of assignment, and nonlinearities/interactions improve explanatory power.

Economic message: Higher-level approvals are not random. They are assigned by observable credit-policy rules. This creates both an identification challenge and the source of plausibly exogenous residual variation.

Table 1
Summary statistics

	Level 1	Level 2	Level 3	Level 4	Level 5
Number of firms	113	158	54	57	42
Percentage of firms	26.7%	37.3%	12.7%	13.4%	9.9%

Panel A: Means of variables by approval level					
Approval level rule variables					
Maturity/collateral score	9.61	11.89	14.37	14.42	16.40
Applicant size (in million \$)	6.06	17.15	17.20	39.05	36.98
Central bank credit score	1.06	1.09	1.24	1.23	1.88
Foreign bank branch guarantee	0.77	0.02	0.01	0.00	0.00
Family company?	0.06	0.07	0.44	0.25	0.29
Product market industry benchmark	0.13	0.10	0.50	0.61	0.50
Financial risk industry benchmark	0.02	0.05	0.35	0.40	0.33
Declining industry	0.04	0.02	0.24	0.09	0.00
Other variables					
Total facilities (in million \$)	5.49	16.26	14.30	34.59	26.41
Total Outstanding (in million \$)	3.03	10.21	8.31	22.24	20.94
Net sales (in million \$)	57.64	140.41	304.90	488.29	545.68
Net income (in million \$)	0.56	1.12	14.66	14.24	55.50
Total assets (in million \$)	64.83	145.87	293.21	862.43	1333.80

Panel B: Mean and standard deviation of information variables by approval level					
Mean					
Objective index	-0.37	-0.19	0.36	0.66	0.34
Performance index	0.01	-0.08	0.35	-0.03	-0.23
Size index	-0.56	-0.21	0.17	1.00	0.72
Subjective index	-0.32	-0.09	0.02	0.57	0.41
Strong subjective index	-0.32	-0.14	0.13	0.36	0.48
Weak subjective index	-0.25	-0.15	0.14	0.40	0.44
Standard deviation					
Objective index	0.85	1.04	0.78	0.76	1.12
Subjective index	0.83	0.97	0.87	1.07	1.16
Objective index (demeaned at loan officer level)	0.86	1.01	0.71	0.65	1.08
Subjective index (demeaned at loan officer level)	0.85	0.94	0.82	0.84	0.97

The table presents summary statistics for the main variables used in the paper by a loan's approval level. Approval level rule variables determine how firms are allocated across approval levels (Levels 1–5) by the bank. Panel A shows the eight primary approval level rule variables. There are an additional 11 rule variables that are used occasionally. The complete list (and definitions) of rule variables is given in Appendix A. Panel B provides summary stats on information indices that are constructed using information collected by a loan officer on an applicant firm. The "Objective index" is constructed using a firm's audited financial ratios and size (see Appendix B for details), while the "Subjective index" is constructed using a subjective assessment of various firm attributes (see Appendix C for details).

Table 2
Mapping of rule variables to level of approval assignment

Dependent variable	Approval level					
	OLS				Ordered probit	
	(1)	(2)	(3)	(4)	(5)	(6)
Maturity/collateral score	0.09 (0.01)	0.06 (0.06)	-0.39 (0.20)	0.12 (0.01)	0.12 (0.02)	0.51 (0.14)
Applicant size	0.01 (0.00)	0.03 (0.01)	0.04 (0.01)	0.01 (0.00)	0.01 (0.00)	0.05 (0.01)
Central bank credit score	0.31 (0.10)	1.36 (0.51)	6.71 (2.93)	0.29 (0.10)	0.29 (0.12)	1.29 (0.70)
Foreign bank branch guarantee	-0.02 (0.12)	-0.29 (0.45)		-0.32 (0.11)	0.42 (0.14)	
Family company?	0.32 (0.16)	0.25 (0.15)			0.44 (0.18)	
Product market industry benchmark	0.48 (0.15)	0.30 (0.13)	0.29 (0.13)		0.55 (0.16)	0.36 (0.17)
Financial risk industry benchmark	0.60 (0.17)	0.41 (0.15)	0.35 (0.15)		0.62 (0.18)	0.45 (0.19)
Declining industry	-0.52 (0.23)	-0.67 (0.20)	-0.68 (0.20)		-0.51 (0.25)	-0.81 (0.25)
Pairwise interaction of top 4 rule variables		Yes	Yes			Yes
Powers 2 and 3 of rule variables included?		Yes	Yes			Yes
No. of obs.	424	424	424	424	424	424
Adj. R -sq/Pseudo R -sq	0.42	0.54	0.55	0.34	0.19	0.29

This table predicts approval level based on functions of rule variables used in the credit manuals to assign loan applicant firms to approval levels. The dependent variable is the "Approval level," which varies from 1 to 5. All regressions include the 19 rule variables described in Appendix A, but coefficients on only the eight primary rule variables are shown for brevity. Columns (1)–(4) report ordinary least squares (OLS) estimates, while Columns (5) and (6) report ordered probit estimates. Column (1) includes only the 19 rule variables as controls. Column (2) adds 16 pairwise interactions of the four top rule variables: "Maturity/collateral score," "applicant size," "central bank credit score," and "foreign bank branch guarantee." Column (3) further includes powers 2 and 3 of the 19 rule variables, i.e., a total of 38 additional controls. Column (4) includes only the four top rule variables on the right-hand side. In Columns (3) and (6), "foreign bank branch guarantee"/"family company?" and "family company?," respectively, are dropped due to perfect collinearity. Robust standard errors clustered at the holding company level are reported in parentheses.

(a) Table 1: summary statistics by approval level

(b) Table 2: mapping rule variables to approval level

6.3 Table 3 and Figure 3: information structure and empirical strategy

Read:

- Table 3 reports positive correlations among objective subcomponents, among subjective subcomponents, and between the overall objective and subjective indices.
- The objective and subjective indices are correlated but far from identical, leaving room to estimate their separate roles in approval decisions.
- Figure 3 illustrates the endogenous-assignment problem: a rule variable may be correlated with the informativeness of objective and subjective information.
- The figure also clarifies the identification idea: after controlling for smooth functions of rule variables, nonlinear jumps in approval assignment help identify the hierarchy effect.

Economic message: The paper is not simply comparing high-level and low-level borrowers. It explicitly addresses the possibility that borrowers sent higher up differ in how informative their hard and soft information is.

Table 3
Correlation of matrix of information indices and their subcomponents

Panel A: Correlation matrix for subcomponents of objective index								
	Pretax interest coverage	Pretax funds flow interest coverage	Funds from operations/ total debt	Free oper. cash flow/ total debt	Pretax return on avg. capital	Total debt/ capitalization	Current ratio	Firm size
Pretax interest coverage	1.00							
Pretax funds flow? interest coverage	0.91	1.00						
Funds from oper./ total debt (%)	0.67	0.75	1.00					
Free oper. cash flow/ total debt (%)	0.42	0.44	0.61	1.00				
Pretax return on avg. capital (%)	0.66	0.53	0.62	0.38	1.00			
Total debt/ capitalization (%)	0.40	0.44	0.68	0.47	0.27	1.00		
Current ratio (dec.)	0.20	0.19	0.24	0.14	0.09	0.33	1.00	
Firm size	0.13	0.12	0.04	0.03	0.03	0.03	-0.17	1.00

Panel B: Correlation matrix for subcomponents of subjective index					
	Industry risk assessment	Competitive position	Management quality	Risk management policies	Access to capital
Industry risk assessment	1.00				
Competitive position	0.40	1.00			
Management quality	0.44	0.67	1.00		
Risk management policies	0.41	0.54	0.61	1.00	
Access to capital	0.49	0.64	0.67	0.51	1.00

Panel C: Correlation matrix for information indices						
	Objective index	Performance index	Size rating	Subjective index	Strong subjective index	Weak subjective index
Objective index	1.00					
Performance index	0.73	1.00				
Size rating	0.73	0.06	1.00			
Subjective index	0.46	0.23	0.44	1.00		
Strong subjective index	0.42	0.20	0.42	0.78	1.00	
Weak subjective index	0.43	0.22	0.40	0.79	0.77	1.00

This table reports the correlation between information indices and their subcomponents. See Appendix B and Appendix C for variable description and summary statistics for these variables.

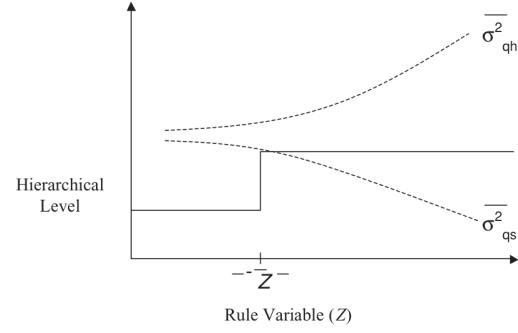


Figure 3
Empirical strategy

(a) Table 3: correlations among information indices

(b) Figure 3: nonlinear rule-based identification

6.4 Table 4 and Table 5: main hierarchy effect and geographical distance

Read:

- Table 4 is the paper's central result. At high approval levels, sensitivity to subjective information falls and sensitivity to objective information rises.
- In the simpler specification, the subjective-index by high-level interaction is negative, while the objective-index by high-level interaction is positive.
- With loan-officer fixed effects and rule-variable controls, the same pattern remains: subjective information receives less weight higher in the hierarchy, and objective information receives more weight.
- Opening the five levels shows that the change is not gradual. It appears sharply around Level 3.
- Table 5 shows why Level 3 matters: this is where geographical separation begins. The effect is strongest when the approving officer is in a different branch from the information-collecting loan officer.

Economic message: Hierarchy matters because it separates the information collector from the decision maker. The geography result makes the communication interpretation especially persuasive.

Table 4
Does reliance of information vary with hierarchical distance?

Dependent variable	Log(approved credit)					
	(1)	(2)	(3)	(4)	(5)	(6)
High level	0.53 (0.15)	0.32 (0.19)	0.15 (0.26)	0.17 (0.27)		
Subjective index	0.41 (0.08)	—	—	—	—	—
Objective index	−0.04 (0.06)	—	—	—	—	—
Subjective index × high level	−0.43 (0.13)	−0.78 (0.23)	−0.68 (0.31)	−0.60 (0.31)		
Objective index × high level	0.84 (0.13)	0.92 (0.29)	0.82 (0.28)	0.93 (0.31)		
Subjective index × Level 2				−0.12 (0.17)	0.24 (0.33)	
Subjective index × Level 3				−0.87 (0.31)	−0.76 (0.40)	
Subjective index × Level 4				−0.91 (0.36)	−0.62 (0.42)	
Subjective index × Level 5				−0.88 (0.48)	−0.74 (0.53)	
Objective index × Level 2				0.25 (0.14)	−0.25 (0.38)	
Objective index × Level 3				1.13 (0.40)	0.92 (0.42)	
Objective index × Level 4				1.37 (0.39)	0.70 (0.46)	
Objective index × Level 5				1.14 (0.38)	0.57 (0.37)	
Constant	1.72 (0.07)		—	—	—	
Rule variables and their information interactions			Yes	Yes	Yes	Yes
Loan officer FE and their information interactions		Yes	Yes	Yes	Yes	Yes
Industry FE, firm size, and ROA				Yes		
Indicator variables for each level				Yes	Yes	
No. of obs.	424	424	424	423	424	424
Adj. <i>R</i> -sq	0.27	0.39	0.65	0.73	0.43	0.66
Adj. <i>R</i> -sq	0.27	0.39	0.65	0.73	0.43	0.66

This table estimates the credit sensitivity to objective and subjective information variables for firms getting credit approvals at various hierarchical levels within the bank. The dependent variable is the logarithm of the amount of credit approved, log(approved credit). "High level" is an indicator variable that collapses the 5 approval levels into "high" and "low" around the median. "High level" takes a value of 1 for Levels 3–5; and a value of 0 otherwise. Columns (2)–(6) add loan office fixed effects and their interactions with the subjective and objective information variables (75 additional controls in all). Columns (3), (4), and (6) further add the 19 rule variables used by the bank to assign firms to different levels and their interactions with objective and subjective information measures (57 controls in all). The 19 rule variables are described in Appendix A. Columns (5) and (6) open up the five hierarchical levels to see how sensitivity to information changes at each level. The omitted category in these columns is "Level 1." Robust standard errors clustered at the holding company level are reported in parentheses.

Table 5
Geographical distance

Panel A: Joint distribution of hierarchical and geographical distance						
Geographical distance	Level of approval (Hierarchical distance)					
	1	2	3	4	5	
0	113	158	17	0	0	
1	0	0	37	57	42	

Panel B: Information use and geographical distance			
Dependent variable	Log(approved credit) Level 3 only		
	(1)	(2)	(3)
Geographical distance × objective index	0.86 (0.13)	0.89 (0.45)	0.89 (0.42)
Geographical distance × subjective index	−0.44 (0.14)	−0.36 (0.38)	−0.36 (0.36)
Other variables included in regression but coefficients not shown	*	*	**
No. of obs.	424	54	424
Adj. <i>R</i> -sq.	0.27	0.28	0.34

Panel A reports the joint distribution of the level of approval, and geographical distance between a loan officer and the ultimate officer approving a loan. "Geographical distance" takes a value of 0 if the loan officer who collects information and the ultimate approving officer are located in the same branch; otherwise it is coded as 1. Panel B estimates the credit sensitivity to objective and subjective information variables for firms getting credit approvals from officers located at different facilities. The dependent variable is the logarithm of the amount of approved credit, log(approved credit). *Geographical distance, objective index, subjective index, and a constant. **Geographical distance, objective index, subjective index, approval level indicator variables, and the interaction of these indicator variables with the objective index and the subjective index.

(a) Table 4: information sensitivity by hierarchical distance

(b) Table 5: geographical distance and information use

6.5 Table 6 and Table 7: experience and more subjective information

Read:

- Table 6 asks whether experienced loan officers communicate subjective information more effectively.
- The positive triple interaction for subjective information, high level, and tenure means that experience mitigates the decline in subjective-information use at higher levels.
- The negative triple interaction for objective information is consistent with less need to substitute toward objective information when subjective information is more credible.
- Table 7 decomposes subjective information into strong and weak subjective components.
- The stronger decline is for the strong subjective index, which includes more judgment-based information such as management quality and competitive position.

Economic message: The paper's mechanism is strongest exactly where it should be strongest: for more subjective information and for relationships with less accumulated trust or shared experience.

Table 6
Does experience help the use of subjective information?

Dependent variable	Log(approved credit)	
	(1)	(2)
Subjective index $\times$ high level $\times$ tenure	2.16 (0.56)	2.83 (0.47)
Objective index $\times$ high level $\times$ tenure	-1.16 (0.65)	-1.70 (0.49)
Rule variables and their information interactions		Yes
Loan officer FE and their information interactions	Yes	Yes
Other variables included in regression but coefficients not shown	*	*
No. of obs.	424	424
R-sq.	0.40	0.66

This table tests whether the experience of a loan officer who collects information helps in the use of subjective information in the credit approval decision. "Tenure" is a dummy variable that takes a value of 1 when the years of experience in the bank are seven or more years, and 0 otherwise. The dependent variable is the logarithm of the amount of approved credit, log(approved credit). Column (1) adds loan office fixed effects and their interactions with subjective and objective information variables (75 additional controls in all). Column (2) further adds the 19 rule variables used by the bank to assign firms to different levels and their interactions with objective and subjective information measures (57 controls in all). The 19 rule variables are described in Appendix A. Robust standard errors clustered at the holding company level are reported in parentheses. *Subjective index, objective index, high level, subjective index $\times$ high level, objective index $\times$ high level, tenure, tenure $\times$ subjective index, tenure $\times$ objective index, and tenure $\times$ high level.

(a) Table 6: loan-officer experience and subjective information

Table 7
Decomposing subjective information

Dependent variable	Log(approved credit)				
	(1)	(2)	(3)	(4)	(5)
Subjective index $\times$ high level	-0.68 (0.19)	-0.65 (0.19)			
Objective index $\times$ high level	0.85 (0.25)	0.84 (0.25)	0.90 (0.13)	0.87 (0.27)	0.85 (0.25)
Weak subjective index $\times$ high level			-0.18 (0.18)	-0.29 (0.21)	-0.23 (0.20)
Strong subjective index $\times$ high level			-0.40 (0.19)	-0.58 (0.20)	-0.52 (0.19)
Definition of subjective rating	Average	Weighted			
Rule variables with information interactions	Yes	Yes			Yes
Loan officer FE and information interactions	Yes	Yes		Yes	Yes
Other variables included in regression but coefficients not shown	*	*	**	**	**
No. of obs.	409	409	409	409	409
Adj. R-sq.	0.73	0.73	0.29	0.42	0.74

This table checks the robustness of the main results by using different measures of the subjective index. We construct two different measures of overall subjective information rank. AVG subjective is a simple arithmetic mean of all the 18 subjective information variables. WAVG subjective weighs the five categories that form the subjective index equally, while giving equal weights to the subjective information variables within each category. Weak subjective index is a standardized average of the categories management quality and competitive position. Strong subjective index is a standardized average of the categories industry risk assessment, risk management policies, and access to capital. Columns (1), (2), (4), and (5) include loan office fixed effects and their interactions with subjective and objective information variables (75 additional controls in all). The dependent variable is the logarithm of the amount of approved credit, log(approved credit). Columns (1), (2), and (5) further add the 19 rule variables used by the bank to assign firms to different levels and their interactions with objective and subjective information measures (57 controls in all). The 19 rule variables are described in Appendix A. Robust standard errors clustered at the holding company level are reported in parentheses. *Subjective index, objective index, and high level. **Objective index, weak subjective index, strong subjective index, high level, and a constant.

(b) Table 7: decomposing subjective information

6.6 Appendix B and Appendix C: construction of objective and subjective variables

Read:

- Appendix B lists the raw objective variables and their implied ratings. These include leverage, profitability, cash-flow, liquidity, and size measures.
- Appendix C lists subjective variables in the five categories used by the bank: industry risk assessment, competitive position, management quality, risk management policies, and access to capital.
- These appendices show that the information measures are detailed and operational, not ad hoc labels.

Economic message: The measurement is central to the contribution. The paper can test organizational theories because it observes separate hard and soft information inputs inside the bank.

Appendix B

Objective information variables

Objective information variable	Mean	SD	Min.	Max.	Obs.
Subcomponent of objective index (raw form)					
Performance index					
Pretax interest coverage (dec.)	4.68	8.12	-8.47	23.62	394
Pretax funds flow interest coverage (dec.)	7.79	10.71	-7.02	30.90	394
Funds from operations/total debt (%)	12.18	35.55	-0.73	121.37	394
Free oper. cash flow/total debt	5.12	14.92	-1.24	50.38	394
Pretax return on avg. capital (%)	0.06	0.24	-0.59	0.54	394
Total debt/capitalization (%)	0.42	0.28	0.00	0.92	394
Current ratio (dec.)	1.25	0.65	0.32	2.86	394
Size index					
Firm size	411,250	4,102,483	-102,675	84,000,000	424
Subcomponent of objective index (implied rating)					
Performance index					
Pretax interest coverage	10.98	8.01	0.00	22.00	394
Pretax funds flow interest coverage	11.39	7.71	0.00	22.00	394
Funds from operations/total debt	10.21	7.86	0.00	22.00	394
Free oper. cash flow/total debt	10.34	8.66	0.00	22.00	394
Pretax return on avg. capital	9.40	8.68	0.00	22.00	394
Total debt/capitalization	14.17	6.23	0.00	22.00	394
Current ratio	7.14	5.86	0.00	22.00	394
Size index					
Firm size	2.26	1.43	1.00	6.00	424
Standardized objective indices					
Objective index	0.00	1.00	-41.90	3.22	424
Performance index	0.00	1.00	-1.88	2.47	424
Size index	0.00	1.00	-0.89	2.62	424

This table reports summary statistics of the objective information variables used in the paper. All subcomponents are reported both in raw form and as an implied rating. The bank translates the ratios according to a prespecified formula into a rating that goes from 0 to 22 for the subcomponents of the performance index, and from 1 to 6 for the component of the size index. For 30 firms, the subcomponents of the performance index measure are missing; hence the total number of firms is 394. All measures are in book values. Pretax interest coverage is the ratio of (net income from operations - gross interest expense - income tax - net sale of equity)/gross interest expense. Pretax funds flow interest coverage is the ratio of (free cash flow - gross interest expense - income tax - net sale of equity)/gross interest expense. Funds from operations/total debt is the ratio of net income from operations/total debt where total debt is composed by short- and long-term debt. Free operating cash flows/total debt is the ratio of free cash flow/total debt. Pretax return on average capital is the ratio of (net income from operations - gross interest expense - income tax)/(total debt + net worth + minority interest). Total debt/capitalization is the ratio of total debt/(total debt + net worth + minority interest). Current ratio is the ratio of total current assets over total current liabilities. Firm size is the total capitalization of the firm.

(a) Appendix B: objective information variables

Appendix C

Subjective information variables

Subjective information variable	Mean	SD	Min.	Max.	Obs.
Industry risk assessment					
Trend in output	3.51	0.80	1.00	7.00	409
Trend in earnings	3.27	0.78	1.00	7.00	409
Cyclicality	3.35	0.81	1.00	7.00	409
External risks	3.53	0.71	2.00	5.00	409
Mean category	3.41	0.59	1.75	5.00	409
Competitive position					
Market position	4.28	1.47	1.00	7.00	407
Product line diversity	3.88	1.12	1.00	7.00	408
Operating cost advantage	3.46	0.89	1.00	7.00	406
Technology advantage	3.70	0.92	1.00	7.00	406
Key success factors	3.67	0.84	1.00	7.00	406
Mean category	3.80	0.81	1.00	6.60	408
Management quality					
Professionalism	3.67	0.90	1.00	7.00	409
Systems and controls	3.66	0.89	1.00	7.00	409
Financial disclosure	3.72	0.85	1.00	7.00	409
Ability to act decisively	3.77	0.80	1.00	7.00	409
Mean category	3.70	0.75	1.00	6.50	409
Risk management policies					
Leverage policy	3.34	0.85	1.00	7.00	409
Liquidity policy	3.36	0.86	1.00	7.00	409
Hedging policy	3.60	0.86	1.00	7.00	409
Mean category	3.43	0.72	1.00	6.30	409
Access to capital					
Capital markets	3.47	1.11	1.00	7.00	409
Banks	3.77	1.01	1.00	7.00	409
Mean category	3.62	0.98	1.00	7.00	409
Standardized subjective indices					
Overall subjective index	0.00	1.00	-3.75	2.32	424
Strong subjective index	0.00	1.00	-3.85	3.58	409
Weak subjective index	0.00	1.00	-3.35	3.56	409

This table reports summary statistics of the subjective information variables, the primary components, and the subcomponents used in the paper (reported in Appendix D). There are 18 primary subjective information variables, divided across five subjective information categories. The main categories of the subjective index are: "industry risk assessment," "competitive position," "management quality," "risk management policies," and "access to capital." All ratings are reported by the loan officer and are between 1 (worst) and 7 (best). The "mean category" is the average across the fields for each of the categories. There are 15 firms that have the subcomponent level information missing, therefore reducing the sample to 409 firms. Furthermore, for three

(b) Appendix C: subjective information variables

7 Short summary

- The paper studies how hierarchical distance affects the use of information in loan approval.
- The setting is a multinational bank in Argentina with 424 corporate loan applications in 1998.
- Loan officers collect both objective and subjective information, while approval can occur at one of five hierarchical levels.
- Objective information comes from audited financials and size; subjective information comes from loan-officer assessments.
- The main prediction is that subjective information becomes harder to use when the decision maker is farther away from the information collector.
- The main result supports the prediction: higher-level approval reduces sensitivity to subjective information and increases sensitivity to objective information.
- The result is robust to loan-officer fixed effects, rule-variable controls, and interactions between rule variables and information measures.
- The hierarchy effect appears sharply when the approving officer is geographically separated from the loan officer.
- More experienced loan officers partly mitigate the loss of subjective information.
- The effect is stronger for more subjective components of the subjective index.
- The paper's main contribution is to show directly, inside a firm, that hierarchy changes information use.

8 Conclusion

8.1 Core conclusion

Liberti and Mian show that hierarchies affect the way organizations use information. When loan approval is made farther away from the loan officer who collects information, the bank relies less on subjective information and more on objective information. The central conclusion is not just that hierarchy changes who has authority, but that hierarchy changes which information enters the decision.

8.2 Economic intuition

Subjective information is valuable because it captures local knowledge, impressions of management, competitive position, and other nuanced assessments. But this information is difficult to verify and communicate. When the decision maker is close to the loan officer, the approving officer can better interpret, question, and trust the subjective assessment. When the decision maker is farther away, the organization substitutes toward objective, verifiable information.

8.3 Incremental contribution revisited

The paper advances the organizational economics literature by taking theories about information transmission inside firms to detailed intrafirm data. It observes the information collector, the final decision maker, the information used, the hierarchical distance between them, and the loan outcome. This allows the authors to test a core prediction of organizational theory with unusually direct evidence.

8.4 Implications for organizations and finance

The findings imply that organizational design affects not only costs and control but also the information content of decisions. Large banks and other hierarchical organizations may lose valuable soft information as decisions move upward or across locations. At the same time, centralization may be useful for complex loans or risk control. The trade-off is therefore not simply decentralization versus control; it is also soft-information use versus objective-information reliance.

8.5 Limitations

The analysis is based on one bank in one country and one year, so external validity should be evaluated carefully. The data include approved corporate loan applications, not necessarily rejected applications. The paper also cannot fully distinguish among communication costs, incentive effects, and strategic information manipulation, because these mechanisms make similar predictions. Still, the geographical-distance, loan-officer-experience, and strong-subjectivity tests provide strong support for the communication and human-touch interpretation.

8.6 Takeaway

The enduring takeaway is that soft information does not automatically travel inside firms. Hierarchical and geographical distance change how decision makers weight information. Objective information can move through the hierarchy relatively intact, but subjective information depends on proximity, trust, repeated interaction, and shared context.